BOTANY HUB MULTAI

B.Sc. I BOTANY — MINOR PAPER II

DIVERSITY OF PLANTS

UNIT V — Angiosperms, Bentham & Hooker, Malvaceae and Poaceae

DETAILED TEACHER-STYLE NOTES

Definitions • Complete explanations • Classification tables • Life cycles • Flowcharts • Diagrams • Examples • Economic importance • Exam points

Prepared strictly from the uploaded 2025–26 Minor Paper II syllabus.

UNIT V — ANGIOSPERMS

The syllabus covers semi-technical terminology, outline classification by Bentham and Hooker, dicot family Malvaceae and monocot family Poaceae, with herbarium preparation activity. ■filecite■turn11file0■L78-L85■

1. Semi-technical terminology for plant description

Botanical descriptions use standardized terms so that a plant can be recognized precisely. A family description normally follows a fixed sequence: habit → root → stem → leaves → inflorescence → flower → calyx → corolla → androecium → gynoecium → fruit → seed → floral formula → floral diagram where required.

Character	Important terms to know
Habit	Herb, shrub, tree, climber
Leaf arrangement	Alternate, opposite, whorled
Leaf type	Simple, compound
Venation	Reticulate, parallel
Flower symmetry	Actinomorphic, zygomorphic
Sexuality	Bisexual, unisexual
Ovary position	Superior, inferior
Placentation	Axile, parietal, free-central, basal

2. Bentham and Hooker classification — outline

Bentham and Hooker proposed a widely taught natural classification based largely on observable morphological characters. For B.Sc. answers, present the broad sequence rather than memorizing every historical detail without understanding.

Very broad textbook outline

Phanerogams / seed plants → Dicotyledons → Monocotyledons → Gymnosperms in historical treatment

Dicot subdivision in traditional outline	Main idea
Polypetalae	Petals generally free
Gamopetalae	Petals generally united
Monochlamydeae	Single perianth/undifferentiated floral envelope in traditional grouping

3. Family Malvaceae — detailed description

Definition — Malvaceae: A primarily dicot family of herbs, shrubs or trees, commonly characterized in standard textbook examples by mucilage, often stellate hairs, and numerous stamens that may be united into a staminal tube.

Character	Typical Malvaceae description
Habit	Herbs, shrubs or trees
Leaves	Usually alternate, simple, stipulate; often palmate venation
Inflorescence	Usually solitary or cymose in many examples
Flower	Bracteate, bisexual, actinomorphic, usually pentamerous
Calyx	5, gamosepalous; epicalyx may be present in Hibiscus
Corolla	5, polypetalous, twisted/contorted aestivation in common examples
Androecium	Numerous; filaments often united into a staminal tube
Gynoecium	Usually 5 or more carpels, syncarpous; ovary commonly superior
Fruit	Capsule or schizocarp depending on genus

Character	Typical Malvaceae description
Examples	Hibiscus, Gossypium, Abelmoschus

4. Family Poaceae — detailed description

Definition — Poaceae: The grass family, a major monocot family characterized by jointed culms, narrow leaves with sheathing bases, spikelets and highly reduced florets adapted largely for wind pollination.

Character	Typical Poaceae description
Habit	Mostly herbs; grasses and bamboos
Root	Fibrous
Stem	Culm, often cylindrical with nodes and internodes
Leaves	Alternate, narrow, parallel venation; sheath, ligule and blade
Inflorescence	Usually composed of spikelets
Flower/floret	Small, often enclosed by bracts such as lemma and palea
Perianth	Reduced to lodicules
Androecium	Usually 3 stamens
Gynoecium	Usually bicarpellary; ovary superior, unilocular with one ovule
Fruit	Caryopsis
Examples	Wheat, rice, maize, bamboo, sugarcane

Malvaceae vs Poaceae

Feature	Malvaceae	Poaceae
Class	Dicot	Monocot
Root	Usually tap root	Fibrous
Venation	Often reticulate/palmate	Parallel
Flower	Relatively large and conspicuous	Small, reduced floret
Pollination	Often insect-related in common examples	Commonly wind
Fruit	Capsule/schizocarp depending genus	Caryopsis
Economic role	Cotton, vegetables, ornamentals	Major cereals, fodder, sugarcane

5. Herbarium preparation activity

A herbarium specimen is a carefully collected, pressed, dried, mounted and labelled plant sample used for study and reference. The label should include scientific name (if correctly identified), family, locality, date, collector, habitat and diagnostic notes. Never collect protected or threatened species without authorization.

Basic herbarium workflow

Collect ethically → Press → Dry → Mount → Label → Identify → Store

Exam focus: Unit V scoring strategy: write the family description in fixed sequence and underline diagnostic characters. Malvaceae and Poaceae are compulsory family targets in this syllabus. ■filecite■turn11file0■L78-L85■